

PROJECT RISKS AND MITIGATION STRATEGIES

Risk	Description	Mitigation Strategy
Availability	System depends on external APIs. API downtime, pricing changes, usage limits could lead to disruptions.	Service layers for LLM provider Hugging Face is abstracted, allowing fallback to demo mode.
Scalability	Concurrent User requests may overload the backend	System uses asynchronous request handling and container-based system offers horizontal scaling with added orchestration
Security	API keys, SSH keys, and secret data in environment variables can be exposed through misconfiguration	Secret data is stored on .env files locally, nginx routing is used to restrict access via HTTPS
Model Bias	LLMs can produce misleading outputs	Each model response is clearly labelled, allowing responses to be compared for differences as opposed to accuracy
Complexity	The multi-container architecture can easily lead to misconfiguration and deployment failure	Docker compose defines and manages the containers which allows for reproducible steps